



# BPM Adoption in Serbian Companies

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**Abstract.** This paper analysis BPM adoption in context of companies operating in Serbia. The goals were to determine level of BPM adoption in companies in Serbia and identify what factors contributing the most to the success in BPM adoption. Questionnaire was used for data collection. BPM adoption was measured through Process Performance Index (PPI). Parametric statistical tests were used on survey data to identify factors that contribute the most to successful BPM adoption. Results in this research shows several factors that significantly contribute to success of BPM adoption measured through PPI. First, companies who have formally trained their employees in process analysis/redesign experienced more significant success in BPM adoption. Second factor is strategic orientation, expressed through criteria such as enterprise-wide business process architecture design and change efforts, increased market share and revenue, reduction of business risk, etc. This shows that strategic focus can even make up for the lack of operational knowledge, and increase chances for success of BPM adoption.

**Keywords:** BPM adoption · Process Performance Index · BPM practice

## 1 Introduction

Business Process Management (BPM) plays an important role for maintaining efficiency and effectiveness of the operations of companies and organizations [6]. Empirical research suggests positive correlation between BPM and business success [8], and have shown that organization can benefit from BPM through better financial and nonfinancial performance which can drive to competitive advantage [12].

Considering the influence BPM implementation might have on gaining competitive advantage, BPM stands out as a viable solution for improving company performance in transition economies [12]. Recent investigations [4] confirmed the impact of business process orientation on organizational performance in Slovenia and Croatia. Therefore, it is interesting to analyze current level of BPM adoption in Serbia, as transitional economy, which is the main purpose of this paper. Bandara et al. [2] define BPM success as “the resulting status of when the intended goals of the BPM initiative are met to a satisfactory level”. Although BPM is often used for improving organization’s operational competitiveness, research has shown that many BPM projects have been unsuccessful in practice [1, 13], which is why it is important to consider key success factors for BPM adoption. Relevant literature suggests several ways of measuring the

success of BPM adoption [5, 10, 11]. Process Performance Index (PPI), created by Rummler-Brache Group [10] is empirically validated, quantitative, and publicly available [5], making it suitable as BPM adoption measuring instrument. PPI consists of 10 criteria: Alignment with strategy, Holistic approach, Process awareness by management and employees, Portfolio of process management initiatives, Business process improvement (BPI) methodology, Process metrics, Customer focus, Process management, Information systems, and Change management. It offers good balance between overall measure of BPM adoption and more detailed analysis of criteria that define the success of BPM adoption, which is why it has been chosen as a primary metric for this research.

## 2 Research Methodology

The purpose of this study is to determine level of BPM adoption in companies in Serbia measured through PPI, to see what factors contribute the most to high level of PPI and drivers and challenges encountered in BPM practice in Serbia. Survey research was conducted, with questions adapted from BPTrends survey [3], Process Excellence Network survey [9], and Marshall [7]. The questions required single or multiple choices, while some of them included an open form. For each company, total PPI score is calculated, as a sum of ratings by criteria. The data were gathered during March and April of 2019. Questionnaires were sent to 600 companies in Serbia. Total of 61 responses was received, representing 10.16% response rate. All of the responses were included in the results presented in this paper.

## 3 Results

Most companies described the scope of BPI initiatives as enterprise wide (44.3%). The need to improve productivity/efficiency (49 companies), The need to save money (40), The need to improve business coordination and control (30), and The need to improve customer satisfaction (27) are identified as most significant drivers, while Increase in market share and revenue (11) and One time event (4) are least significant. Business process modeling/documenting (46), Business process management (44) are widely implemented, while initiatives that companies plan to implement are related to Core process redesign (23) and Process analysis/redesign training (24). Companies usually face challenges regarding Lack of interest within top management (22) and Absence of skills (18) and Obstacles in new technology implementation (17).

The PPI values range from 10 to 50 for each company. PPI value can be considered moderate (mostly between 30 and 38), meaning success of BPM adoption in companies is moderate. PPI criteria with highest average scores are Holistic approach (3.97) and Process metrics (3.67), while criteria with lowest average scores are Customer focus (3.44), BPI methodology (3.38), and Portfolio of process management initiatives (3.25). Holistic approach, Process management, and Alignment with strategy have relatively low standard deviation, while Process awareness by management and

employees, Customer focus, and BPI methodology have relatively high standard deviation.

Participants were divided into four groups based on the scope of their BPI initiative (1: The managers are trained, but there are no formal programs; 2: Small-scale pilot projects; 3: One or more business units, 4: Enterprise wide). Scores for four groups were statistically different at level  $p < .05$ . Post-hoc comparisons revealed that companies without formal BPI initiative have smaller PPI than companies with programs in one or more business units or preferably on enterprise level.

Results of t-test show that companies oriented more towards market share and revenue have higher PPI ( $t(59) = -2.157$ ,  $p = 0.035$ ). The magnitude of differences in means ( $MD = -5.196$  95%  $CI$ :  $-10.017$  to  $-0.376$ ) is moderate ( $\eta^2 = .07$ ). Companies oriented more towards reducing business risk have higher PPI ( $t(59) = -1.975$ ,  $p = 0.05$ ), with the effect that can be considered moderate ( $MD = -4.274$  95%  $CI$ :  $-8.603$  to  $0.055$ ,  $\eta^2 = .06$ ). Companies that worked on business process architecture development initiatives have higher PPI ( $t(59) = -2.780$ ,  $p = 0.007$ ) with the effect closed to large ( $MD = -5.030$  95%  $CI$ :  $-8.650$  to  $-1.410$ ,  $\eta^2 = .12$ ). Companies that worked on Process change on enterprise level have also higher PPI ( $t(59) = -2.662$ ,  $p = 0.01$ ), with moderate magnitude of differences in means ( $MD = -4.919$  95%  $CI$ :  $-8.617$  to  $-1.221$ ,  $\eta^2 = .10$ ).

Mann-Whitney U test revealed that companies with trained employees in Process analysis/redesign have succeeded in identifying and eliminating problems and disagreement ( $Md = 4.00$ ,  $N = 15$ ),  $U = 195.000$ ,  $z = -2.699$ ,  $p = 0.00$ ,  $r = 0.35$ ) more than companies without trained employees ( $Md = 3.00$ ,  $N = 46$ ).

## 4 Discussion and Conclusion

There are certain discrepancies between main drivers behind process change and business process initiatives implemented in companies. While drivers are oriented more towards improving operational performance, initiatives are mainly focused on high-level transformation of the company. This points to misalignment of goals and tools used to achieve those goals. Misalignment is also evident with drivers behind process change initiatives and the effects of these initiatives as measured by PPI criteria. For example, customer satisfaction is recognized as one of the main drivers behind process change initiatives, while customer focus is one of the PPI criteria with the lowest score. Lack of knowledge regarding process initiatives might be the cause of this, calling for more systematic education regarding options managers have at hand. The results back this claim up, as companies that undertook more serious process analysis/redesign training had more success in identifying and eliminating problems and inconsistencies. PPI criteria such as alignment with strategy and holistic approach to BPM show high level of agreement among respondents, meaning that strategic orientation of BPM is recognized among companies in Serbia. However, the results show lack of systematic approach to business process initiatives. For example, criteria such as Portfolio of process management initiatives or the existence of formal BPI methodology show more dispersion in the answers. These results are in line with comments expressed earlier,

confirming misalignment between goals and ways to operationalize those goals, which means that both education and the development of formal approach to BPM are needed.

The results show that strategic orientation towards BPM significantly contributes to success of BPM adoption measured through PPI. This is in line with results showing that companies who have formally trained their employees in process analysis/redesign experienced more significant success in BPM adoption. This is expressed through drivers such as increased market share and revenue, and reduction of business risk, and by decision to perform enterprise wide BPI initiatives such as business process architecture development and enterprise level process change management. This shows the importance of process management efforts on an enterprise level, rather than through isolated process improvement initiatives. In addition, it shows that strategic focus can even make up for the lack of operational knowledge, and increase chances for success of BPM adoption.

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